

Item	Value
Project	PoultryCockpit
Document	Functional Specification
Version	1.0
Status	APPROVED FOR DEVELOPMENT
Owner	Sangeeth K
Prepared By	Sangeeth K & ChatGPT
Baseline Date	<i>(Approval Date)</i>
Total Chapters	22
Change Control	Any modification to approved business rules, workflows or calculations shall require a new document version.

Chapter 1 — Project Overview

1.1 Purpose

PoultryCockpit is a desktop-first Progressive Web Application (PWA) for managing commercial broiler poultry farms. It digitizes the complete batch lifecycle, centralizes operational data, automates business calculations, and provides real-time operational and financial insights.

The application is intended to replace spreadsheets and manual calculations with a single, consistent system.

1.2 Primary Objectives

The system shall:

- Manage the complete lifecycle of every poultry batch.
- Record daily farm operations.
- Automate business calculations through a centralized Business Engine.

- Track financial transactions and settlements.
 - Provide operational intelligence and historical analysis.
 - Maintain a complete audit trail of user actions.
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1.3 Core Principles

The following principles apply throughout the application.

Business Engine First

All calculations, formulas, validations, KPIs, and intelligence must be executed by the Business Engine.

The UI is responsible only for capturing input and displaying results.

Single Source of Truth

Business data shall exist only once.

Reports, dashboards, and intelligence modules must consume processed data instead of maintaining independent calculations.

Simple User Experience

The interface should remain clean and task-focused. Complex calculations and business rules must never be exposed to the user.

Modular Architecture

Each functional area shall be implemented as an independent module with clearly defined responsibilities.

Scalability

The architecture should support future expansion without requiring major redesign.

1.4 Application Modules

The system consists of the following modules:

- Dashboard
 - Batch Centre
 - Batch Details
 - Daily Log
 - Liftings
 - Financial Ledger
 - Final Settlement
 - Intelligence
 - Reports
 - Settings
 - Business Engine
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1.5 Batch Lifecycle

Every batch shall follow the lifecycle below.

Shed Readiness



Batch Setup



Place Chicks



Running



Daily Operations



Liftings



Financial Ledger



Final Settlement



Archive

A batch may progress only in this sequence unless explicitly permitted by business rules.

1.6 Business Engine

The Business Engine is the core processing layer of PoultryCockpit.

It is responsible for:

- Business calculations
- Formula execution
- Derived metrics
- Validation rules
- Alerts
- KPIs
- Settlement calculations
- Intelligence generation
- Report calculations

Business logic must not be duplicated within UI components.

1.7 Out of Scope (Version 1)

The following features are intentionally excluded:

- Layer farm management
 - Hatchery management
 - Feed manufacturing
 - Payroll
 - Vehicle logistics
 - Procurement management
 - Offline synchronization
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1.8 Development Guidelines

The following rules are mandatory throughout development.

- Implement business logic only within the Business Engine.
- Keep UI components presentation-focused.
- Avoid duplicate calculations.
- Avoid duplicate master data.
- Ensure all significant user actions are auditable.
- Prefer reusable implementations over duplicated code.
- Follow the approved UI without introducing additional screens unless explicitly approved.

Chapter 2 – System Architecture

Section	Specification
Purpose	Define the overall architecture of PoultryCockpit, establish module boundaries, and specify where business logic, UI logic, and data processing must reside.
Architecture Style	Modular, layered, desktop-first Progressive Web Application (PWA). Each module is independently maintainable while sharing common services and the centralized Business Engine.
Technology Stack	Frontend: Next.js (App Router), React, TypeScript, Tailwind CSS. Backend: Supabase (PostgreSQL). Hosting: Vercel. Database: PostgreSQL. Architecture: Service-oriented with Business Engine separation.
Application Layers	Presentation Layer → UI Screens and Components. Service Layer → Data access, CRUD operations, API communication. Business Engine → Calculations, validations, KPIs, derived metrics, settlement logic, intelligence generation. Database Layer → Persistent storage.
Presentation Layer Responsibilities	Display data, collect user input, perform basic client-side validation (required fields, formats), navigate between screens, and render Business Engine results. No business calculations are permitted.
Service Layer Responsibilities	Handle CRUD operations, communicate with Supabase, transform data where necessary, and invoke Business Engine services. Must not contain business rules.
Business Engine Responsibilities	Execute all calculations, business rules, validations, KPIs, financial computations, settlement calculations, intelligence generation, alerts, and derived metrics. This is the only layer where business logic is allowed.
Database Responsibilities	Persist application data, maintain referential integrity, enforce constraints, store configuration and historical records, and support audit logging.
Core Modules	Dashboard, Batch Centre, Batch Details, Daily Log, Liftings, Financial Ledger, Final Settlement, Intelligence, Reports, Settings, Business Engine.
Shared Components	Navigation, Search, Filters, Dialogs, Tables, Forms, Notifications, File Attachments, Export/Import, Authentication (if enabled), Audit Logging.
Business Rule	Every calculation must originate from the Business Engine. UI and database triggers must not duplicate business logic.

Data Flow	User Action → UI → Service Layer → Business Engine (if required) → Database → Service Layer → UI
Module Independence	Modules may share common services and master data but shall not directly manipulate another module's internal logic. Cross-module operations must occur through defined services or the Business Engine.
Error Handling	Validation errors should be displayed in the UI. System errors should be logged and presented with user-friendly messages. Business rule violations must originate from the Business Engine.
Scalability Requirement	New modules should be added without requiring changes to existing module implementations wherever possible.
Audit Requirement	Every significant create, update, archive, restore, and settlement action must be audit logged.
Performance Requirement	Large tables shall support pagination and filtering. Dashboards and reports should use optimized queries and aggregated data where appropriate.
Development Rules	1. Never implement business logic inside UI components. 2. Never duplicate calculations. 3. Prefer reusable services over duplicated implementations. 4. Keep modules loosely coupled. 5. Follow the approved UI exactly unless a change is explicitly approved.

Development Contract

Requirement	Status for Completion
Business logic exists only within the Business Engine	✓ Mandatory
UI contains no business calculations	✓ Mandatory
Services remain reusable and modular	✓ Mandatory
Database stores data but does not implement business rules	✓ Mandatory
Module boundaries are maintained	✓ Mandatory
Approved UI is followed without unauthorized changes	✓ Mandatory

Chapter 3 - Business Engine

ID	Requirement	Specification
BE-001	Purpose	The Business Engine is the only layer responsible for executing business rules, calculations, validations, derived metrics, KPIs, alerts, and settlement logic.
BE-002	Responsibility	The UI shall never perform business calculations. It shall only collect input and display results returned by the Business Engine.
BE-003	Single Source of Truth	Every calculation shall exist in exactly one place. Duplicate implementations are not permitted.
BE-004	Trigger	The Business Engine shall execute automatically whenever business data is created, updated, archived, restored, or deleted.
BE-005	Processing	All calculations shall be deterministic. Given the same inputs, the engine shall always produce identical outputs.
BE-006	Configuration	Wherever possible, calculations shall be configurable through Settings rather than hardcoded.
BE-007	Validation	Business validations shall execute before data is committed to the database. Invalid transactions shall be rejected with meaningful error messages.
BE-008	Derived Values	Values such as mortality %, feed per bird, water per bird, production cost, FCR, GC, KPI scores, etc., shall be calculated by the Business Engine and never manually entered.
BE-009	Recalculation	Whenever historical data changes, all dependent calculations shall be recalculated automatically.
BE-010	Logging	Every calculation failure shall be logged with sufficient information for debugging.
BE-011	Performance	The engine shall calculate only affected records wherever possible. Full recalculation should be avoided unless explicitly required.
BE-012	Extensibility	New business rules shall be added without modifying existing calculation logic wherever practical.

Business Engine Responsibilities

ID	Function	Responsibility
BEN-001	Batch Calculations	Birds in house, mortality, culls, liftings, closing balance.

BEN-002	Feed Calculations	Feed consumed, feed remaining, feed per bird, cumulative feed.
BEN-003	Water Calculations	Water consumed, water per bird, cumulative water.
BEN-004	Medicine Calculations	Medicine usage summary and history.
BEN-005	Financial Calculations	Expenses, income, production cost, settlement values.
BEN-006	KPI Engine	Daily KPIs, batch KPIs and dashboard metrics.
BEN-007	Intelligence Engine	Trends, comparisons, anomalies and recommendations.
BEN-008	Settlement Engine	Final settlement calculations after batch completion.
BEN-009	Alert Engine	Mortality alerts, low inventory, abnormal consumption and configurable thresholds.
BEN-010	Reporting Engine	Aggregate values required for reports and exports.

Business Engine Rules

ID	Rule
BR-001	Never calculate business values inside UI components.
BR-002	Never duplicate formulas in multiple locations.
BR-003	Never store values that can always be derived unless performance requires caching.
BR-004	Cached values must be automatically refreshed after every relevant transaction.
BR-005	Every calculation must be traceable back to its source data.
BR-006	Business rules must remain independent of UI implementation.

BR-0 Reports, dashboards and intelligence modules must consume Business Engine
07 outputs instead of implementing independent calculations.

Inputs

Source

Batch Details

Daily Log

Liftings

Financial Ledger

Settings

Standard Curves

Lookup Tables

Custom Metrics

Outputs

Destination

Dashboard

Batch Summary

Financial Ledger

Final Settlement

Reports

Intelligence

Notifications

KPIs

Charts

Development Contract

ID	Acceptance Criteria
DC-001	No business calculations exist outside the Business Engine.
DC-002	All derived values originate from the Business Engine.
DC-003	Every calculation is reusable across the application.
DC-004	Every dependent value is recalculated after relevant data changes.
DC-005	Business logic remains independent of UI implementation.

Chapter 4 - Application Rules

ID	Rule Category	Specification
AR-001	Single Active Batch	A shed can have only one active batch at any given time.
AR-002	Batch Lifecycle	Every batch shall follow the approved lifecycle without skipping stages unless explicitly permitted.
AR-003	Data Ownership	Every operational record shall belong to exactly one batch.
AR-004	Historical Integrity	Completed batches shall remain immutable except through authorized administrative actions.
AR-005	Soft Delete	Business records shall be archived instead of permanently deleted wherever possible.
AR-006	Audit Trail	Every significant create, update, archive and restore operation shall be audit logged.
AR-007	Time Consistency	Transactions shall always use the recorded transaction date rather than the current system date for calculations.
AR-008	Master Data Dependency	Operational records shall reference master data instead of storing duplicated values wherever practical.
AR-009	Derived Data	Users shall not manually edit values generated by the Business Engine.
AR-010	Mandatory Data	Required fields shall be validated before saving any transaction.

Batch Rules

ID	Rule	Specification
BAT-001	Batch Creation	A batch can only be created for a shed that is currently available.
BAT-002	Batch Number	Every batch shall have a unique batch identifier.
BAT-003	Batch Status	Valid statuses are Draft, Ready, Running, Completed and Archived.
BAT-004	Running Batch	Only one Running batch is allowed per shed.
BAT-005	Completion	A batch cannot be completed until all birds have been lifted or otherwise accounted for.
BAT-006	Archive	Only completed batches may be archived.
BAT-007	Restore	Restoring an archived batch requires administrator permission.

Daily Log Rules

ID	Rule	Specification
DLG-001	One Entry Per Day	Only one Daily Log is allowed per batch per calendar date.
DLG-002	Chronological Order	Daily Logs shall be maintained in chronological order.
DLG-003	Edit Permission	Existing Daily Logs may be edited only by authorized users.
DLG-004	Future Dates	Future dated Daily Logs are not permitted.
DLG-005	Derived Values	Daily summaries shall be calculated automatically.

Lifting Rules

ID	Rule	Specification
LFT-00 1	Running Batch	Liftings can only be recorded for Running batches.
LFT-00 2	Bird Balance	Total lifted birds shall never exceed birds available.
LFT-00 3	Final Lifting	Final lifting shall automatically reduce Birds in House to zero after reconciliation.
LFT-00 4	Multiple Liftings	Multiple lifting transactions are permitted for a batch.
LFT-00 5	Settlement Dependency	Settlement shall use all recorded lifting transactions.

Financial Rules

ID	Rule	Specification
FIN-0 01	Ledger Entries	Every financial transaction shall belong to one batch.
FIN-0 02	Edit Tracking	Financial modifications shall be audit logged.
FIN-0 03	Derived Totals	Totals shall always be calculated automatically.
FIN-0 04	Currency	All financial values shall use the configured application currency.
FIN-0 05	Settlement Lock	Financial values used for settlement shall not change after settlement approval unless reopened by an authorized user.

Settlement Rules

ID	Rule	Specification
SET-00 1	Prerequisite	Settlement is allowed only after all liftings are complete.

SET-00 2	Calculation	Settlement values shall always originate from the Business Engine.
SET-00 3	Approval	Approved settlements become read-only.
SET-00 4	Reopen	Reopening a settlement requires administrator permission.
SET-00 5	Archive Dependency	Batch archive is allowed only after settlement completion.

Intelligence Rules

ID	Rule	Specification
INT-001	Read Only	Intelligence modules shall never modify business data.
INT-002	Source	Intelligence shall consume Business Engine outputs only.
INT-003	Historical Comparison	Comparisons may include both active and archived batches.
INT-004	KPIs	KPIs shall always reflect the latest processed data.

Reporting Rules

ID	Rule	Specification
REP-00 1	Read Only	Reports shall never modify business data.
REP-00 2	Live Data	Reports shall use the latest available processed data unless explicitly filtered.
REP-00 3	Export	Reports shall support export using the approved export formats.
REP-00 4	Filters	Reports shall respect all applied filters before generation.

Security Rules

ID	Rule	Specification
SEC-001	Authentication	Every user shall be authenticated before accessing protected modules. <i>(Ignore if authentication is disabled in the current deployment.)</i>
SEC-002	Authorization	Users shall access only functions permitted by their assigned role.
SEC-003	Audit	Sensitive operations shall be recorded in the audit log.
SEC-004	Session	Expired sessions shall prevent further data modification.

General Development Rules

ID	Rule
DEV-001	Never hardcode business constants that belong in Settings.
DEV-002	Never duplicate calculations.
DEV-003	Never bypass the Business Engine.
DEV-004	Never manipulate database records directly from UI components.
DEV-005	Keep modules independent wherever possible.
DEV-006	Reuse services instead of duplicating logic.
DEV-007	Prefer configuration over code where applicable.
DEV-008	Follow the approved UI without adding or removing screens unless explicitly approved.

Development Contract

ID	Acceptance Criteria
DC-401	All application rules are enforced consistently across modules.
DC-402	Business rules execute only through the Business Engine.
DC-403	No module contains duplicated business logic.
DC-404	Historical data integrity is preserved.
DC-405	All restricted operations are permission controlled.
DC-406	All significant operations are audit logged.

Chapter 5 - Dashboard

Module Information

Item	Value
Module	Dashboard
Module ID	DASH
Purpose	Display live operational KPIs, active batches, alerts and quick navigation.
Users	Owner, Manager, Supervisor
Data Source	Business Engine
Access	Read Only

Components

ID	Component	Description
DASH-001	Farm Selector	Filters dashboard by farm.
DASH-002	Shed Selector	Filters dashboard by shed.
DASH-003	KPI Cards	Displays live operational metrics.

DASH-004	Active Batches	Lists all running batches.
DASH-005	Alerts	Displays active system alerts.
DASH-006	Quick Actions	Shortcut buttons to major modules.
DASH-007	Charts	Displays operational trends.
DASH-008	Recent Activity	Displays latest business activities.

User Actions

ID	Action	Result
DASH-101	Select Farm	Refresh dashboard.
DASH-102	Select Shed	Filter dashboard.
DASH-103	Open Batch	Navigate to Batch Details.
DASH-104	Open Daily Log	Navigate to Daily Log.
DASH-105	Open Reports	Navigate to Reports.
DASH-106	Open Intelligence	Navigate to Intelligence.
DASH-107	Open Alert	Navigate to related module.

Business Rules

ID	Rule
DASH-201	Dashboard is read-only.
DASH-202	All displayed values shall come from the Business Engine.
DASH-203	Dashboard shall automatically refresh after relevant business transactions.
DASH-204	Only active alerts shall be displayed.
DASH-205	All charts and KPIs shall respect selected Farm and Shed filters.

Validations

ID	Validation
DASH-301	Farm selection is mandatory.
DASH-302	Selected Shed must belong to the selected Farm.
DASH-303	If no data exists, display the standard Empty State.

Navigation

Source	Destination
Dashboard	Batch Centre
Dashboard	Batch Details
Dashboard	Daily Log
Dashboard	Reports
Dashboard	Intelligence
Dashboard	Settings

Dependencies

Dependency	Purpose
Business Engine	KPIs, Alerts, Charts
Batch Centre	Running batch information
Daily Log	Daily operational statistics
Financial Ledger	Financial summaries
Intelligence	Trend analysis

Acceptance Criteria

ID	Requirement
DASH-401	Dashboard loads without errors.
DASH-402	All KPI values match Business Engine outputs.
DASH-403	Filters update all dashboard widgets correctly.
DASH-404	Navigation opens the correct module.
DASH-405	Dashboard performs no business calculations locally.

Chapter 6 - Batch Centre

Module Information

Item	Value
Module	Batch Centre
Module ID	BAT
Purpose	Manage the complete lifecycle of poultry batches.
Users	Owner, Manager, Supervisor
Access	Read, Create, Update, Archive

Business Actions

ID	Action	Description
BAT-001	Create Batch	Create a new batch for an available shed.
BAT-002	Edit Batch	Update batch information before completion.
BAT-003	Archive Batch	Archive a completed batch.
BAT-004	Open Batch	Open Batch Details.

Business Rules

ID	Rule
BAT-101	A shed can have only one active batch.
BAT-102	Batch Number shall be unique.
BAT-103	Only completed batches can be archived.
BAT-104	Archived batches are read-only.
BAT-105	Batch summary values shall be generated by the Business Engine.

Validations

ID	Validation
BAT-201	Farm is mandatory.
BAT-202	Shed is mandatory.
BAT-203	Placement Date is mandatory.
BAT-204	Selected shed must not contain another active batch.
BAT-205	Batch Number must be unique.

Data Sources

Source	Usage
Business Engine	Batch Summary
Daily Log	Live Statistics
Liftings	Bird Balance
Financial Ledger	Financial Summary
Final Settlement	Completion Status

Acceptance Criteria

ID	Requirement
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- BAT-301 Only one active batch exists per shed.
- BAT-302 Batch Number remains unique.
- BAT-303 Archived batches cannot be modified.
- BAT-304 Batch summary values match Business Engine calculations.
- BAT-305 Batch lifecycle follows approved business rules.

Chapter 7 - Batch Details

Module Information

Item	Value
Module	Batch Details
Module ID	BDT
Purpose	Display and manage all operational information for a single batch throughout its lifecycle.
Users	Owner, Manager, Supervisor
Access	Read, Update

Business Actions

ID	Action	Description
BDT-001	Update Batch	Update editable batch information.
BDT-002	Start Batch	Change batch status from Ready to Running.
BDT-003	Complete Batch	Mark batch as completed after settlement prerequisites are met.
BDT-004	View Daily Log	Open Daily Log for this batch.

BDT-005	View Liftings	Open Liftings for this batch.
BDT-006	View Financial Ledger	Open Financial Ledger for this batch.
BDT-007	View Settlement	Open Final Settlement for this batch.

Business Rules

ID	Rule
BDT-101	Batch information may be edited only while permitted by the batch status.
BDT-102	Batch status shall follow the approved lifecycle.
BDT-103	Batch summary values shall always be generated by the Business Engine.
BDT-104	Birds in House shall always equal Opening Birds - Mortality - Culls - Lifted Birds.
BDT-105	A batch cannot be marked as Completed until all birds are accounted for.
BDT-106	Completed batches become read-only except for authorized administrative actions.

Validations

ID	Validation
BDT-201	Batch Number must remain unique.
BDT-202	Placement Date cannot be changed after the batch starts unless authorized.
BDT-203	Birds placed must be greater than zero.
BDT-204	Status changes shall satisfy all business rules.

Data Sources

Source	Usage
Business Engine	Batch KPIs and Summary

Daily Log	Daily Operations
Liftings	Bird Movement
Financial Ledger	Financial Summary
Final Settlement	Settlement Status
Settings	Master Configuration

Acceptance Criteria

ID	Requirement
BDT-301	Batch summary values match Business Engine calculations.
BDT-302	Batch status changes follow the approved lifecycle.
BDT-303	Birds in House is always accurate.
BDT-304	Completed batches cannot be edited by normal users.
BDT-305	All linked modules display data for the selected batch only.

Chapter 8 - Daily Log

Module Information

Item	Value
Module	Daily Log
Module ID	DLG
Purpose	Record all daily operational activities and measurements for a running batch.
Users	Owner, Manager, Supervisor
Access	Read, Create, Update

Business Actions

ID	Action	Description
DLG-001	Add Daily Log	Create the daily operational record for a batch.
DLG-002	Edit Daily Log	Update an existing daily record.
DLG-003	Record Mortality	Record daily mortality and culls.
DLG-004	Record Feed	Record feed received and feed consumed.
DLG-005	Record Water	Record daily water consumption.
DLG-006	Record Medicines	Record medicines, vaccines and supplements administered.
DLG-007	Record Activities	Record daily farm activities, observations and remarks.

Business Rules

ID	Rule
DLG-10 1	Only one Daily Log is allowed per batch per calendar date.
DLG-10 2	Daily Logs can only be created for Running batches.
DLG-10 3	Birds in House shall be automatically recalculated after every Daily Log entry.
DLG-10 4	Feed Balance shall be automatically recalculated after every feed transaction.
DLG-10 5	Mortality, feed, water and medicine summaries shall be generated by the Business Engine.
DLG-10 6	Daily Log history shall remain in chronological order.
DLG-10 7	Historical Daily Logs shall not be deleted; corrections shall be made through updates with audit logging.

Validations

ID	Validation
DLG-201	Log Date is mandatory.

- DLG-202 Duplicate Daily Logs for the same batch and date are not permitted.
- DLG-203 Mortality and culls cannot exceed Birds in House.
- DLG-204 Feed, water and medicine quantities cannot be negative.
- DLG-205 Future dates are not permitted.
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Data Sources

Source	Usage
Batch Details	Batch Information
Business Engine	Daily Calculations
Standard Curves	Performance Comparison
Settings	Master Configuration

Acceptance Criteria

ID	Requirement
DLG-301	Only one Daily Log exists per batch per date.
DLG-302	Birds in House updates correctly after every entry.
DLG-303	Feed Balance updates correctly after every transaction.
DLG-304	Daily summaries match Business Engine calculations.
DLG-305	Every modification is audit logged.
DLG-306	Historical Daily Logs remain accurate and chronological.

Chapter 9 - Liftings

Module Information

Item	Value
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Module	Liftings
Module ID	LFT
Purpose	Record all bird lifting transactions and maintain the live bird balance for a batch.
Users	Owner, Manager, Supervisor
Access	Read, Create, Update

Business Actions

ID	Action	Description
LFT-001	Add Lifting	Record a new lifting transaction.
LFT-002	Edit Lifting	Update an existing lifting record.
LFT-003	Delete Lifting	Remove a lifting record (authorized users only).
LFT-004	Complete Liftings	Confirm that all liftings for the batch are complete.

Business Rules

ID	Rule
LFT-101	Liftings can only be recorded for Running batches.
LFT-102	Multiple liftings are permitted for a batch.
LFT-103	Total lifted birds shall never exceed Birds in House.
LFT-104	Birds in House shall be automatically updated after every lifting.
LFT-105	Lifting transactions shall be processed in chronological order.
LFT-106	Completing liftings does not automatically archive the batch.
LFT-107	Final Settlement can begin only after liftings are confirmed complete.

Validations

ID	Validation
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- LFT-201 Lifting Date is mandatory.
- LFT-202 Number of birds lifted must be greater than zero.
- LFT-203 Total Weight must be greater than zero.
- LFT-204 Lifted birds cannot exceed Birds in House.
- LFT-205 Future dates are not permitted.
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Data Sources

Source	Usage
Batch Details	Current Bird Balance
Business Engine	Bird Balance Calculation
Daily Log	Mortality & Culls
Final Settlement	Settlement Calculations

Acceptance Criteria

ID	Requirement
LFT-301	Bird Balance updates correctly after every lifting.
LFT-302	Total lifted birds never exceed available birds.
LFT-303	Multiple lifting transactions are supported.
LFT-304	Final Settlement cannot proceed until liftings are completed.
LFT-305	All lifting transactions are audit logged.

Chapter 10 - Financial Ledger

Module Information

Item	Value
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Module	Financial Ledger
Module ID	FIN
Purpose	Record and manage all batch-related financial transactions until final settlement.
Users	Owner, Manager, Accountant
Access	Read, Create, Update

Business Actions

ID	Action	Description
FIN-001	Add Transaction	Record a financial transaction.
FIN-002	Edit Transaction	Update an existing transaction.
FIN-003	Delete Transaction	Remove a transaction (authorized users only).
FIN-004	Categorize Transaction	Assign the appropriate expense or income category.

Business Rules

ID	Rule
FIN-101	Every transaction shall belong to a single batch.
FIN-102	Income and expense totals shall be calculated automatically.
FIN-103	Production Cost shall be calculated by the Business Engine.
FIN-104	Ledger totals shall update immediately after every transaction.
FIN-105	Approved settlement values shall not change unless the settlement is reopened.
FIN-106	Every financial modification shall be audit logged.

Validations

ID	Validation
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- FIN-201 Transaction Date is mandatory.
- FIN-202 Category is mandatory.
- FIN-203 Amount must be greater than zero.
- FIN-204 Transaction Date cannot be after the settlement date for a closed batch.
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Data Sources

Source	Usage
Business Engine	Production Cost & Financial Calculations
Batch Details	Batch Information
Final Settlement	Settlement Summary
Settings	Financial Categories & Configuration

Acceptance Criteria

ID	Requirement
FIN-301	Financial totals update correctly after every transaction.
FIN-302	Production Cost matches Business Engine calculations.
FIN-303	Every transaction belongs to a single batch.
FIN-304	Financial history remains accurate after edits.
FIN-305	All financial changes are audit logged.

Chapter 11 - Final Settlement

Module Information

Item	Value
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Module	Final Settlement
Module ID	SET
Purpose	Generate, review and finalize the complete financial settlement for a completed batch.
Users	Owner, Manager, Accountant
Access	Read, Generate, Approve

Business Actions

ID	Action	Description
SET-00 1	Generate Settlement	Generate settlement using the Business Engine.
SET-00 2	Review Settlement	Review settlement before approval.
SET-00 3	Approve Settlement	Lock and approve the settlement.
SET-00 4	Reopen Settlement	Reopen an approved settlement (authorized users only).
SET-00 5	Archive Batch	Archive the batch after settlement completion.

Business Rules

ID	Rule
SET-10 1	Settlement values shall always be generated by the Business Engine.
SET-10 2	Settlement can only be generated after all liftings are completed.
SET-10 3	Settlement shall include all financial transactions for the batch.
SET-10 4	Approved settlements become read-only.

SET-10 5	Reopening a settlement requires administrator authorization.
SET-10 6	A batch can be archived only after settlement approval.
SET-10 7	Reopening a settlement shall automatically unlock the associated batch.

Validations

ID	Validation
SET-20 1	All birds must be accounted for before settlement generation.
SET-20 2	All mandatory financial transactions must be recorded.
SET-20 3	Settlement cannot be approved if calculation errors exist.
SET-20 4	Archive is not permitted until settlement is approved.

Data Sources

Source	Usage
Business Engine	Settlement Calculations
Liftings	Bird Performance
Financial Ledger	Income & Expenses
Batch Details	Batch Summary
Settings	Company & Settlement Configuration

Acceptance Criteria

ID	Requirement
SET-30 1	Settlement values match Business Engine calculations.

SET-30 2	Approved settlements cannot be modified.
SET-30 3	Reopening a settlement restores edit access only to authorized users.
SET-30 4	Batch cannot be archived before settlement approval.
SET-30 5	Settlement includes all eligible financial transactions.

Chapter 12 - Intelligence

Module Information

Item	Value
Module	Intelligence
Module ID	INT
Purpose	Provide operational insights, performance analysis and decision support using historical and live farm data.
Users	Owner, Manager
Access	Read Only

Business Actions

ID	Action	Description
INT-00 1	Compare Batches	Compare performance across batches.
INT-00 2	Analyze Trends	View operational and financial trends.
INT-00 3	Review KPIs	View calculated key performance indicators.
INT-00 4	View Alerts	Review anomalies and recommendations generated by the Business Engine.

Business Rules

ID	Rule
INT-101	Intelligence data shall always be generated by the Business Engine.
INT-102	Intelligence is strictly read-only and shall not modify business data.
INT-103	Comparisons may include both active and archived batches.
INT-104	KPIs shall automatically refresh whenever source data changes.
INT-105	Alerts and recommendations shall be generated from configured business rules.
INT-106	Historical data shall remain unchanged by intelligence processing.

Validations

ID	Validation
INT-20 1	Selected comparison batches must belong to the same farm unless cross-farm comparison is explicitly enabled.
INT-20 2	Comparison requires at least two completed batches.
INT-20 3	KPI calculations shall use the latest processed data.

Data Sources

Source	Usage
Business Engine	KPIs, Trends, Recommendations
Batch Details	Batch Information
Daily Log	Operational Performance
Liftings	Bird Performance
Financial Ledger	Financial Performance
Final Settlement	Final Batch Results

Acceptance Criteria

ID	Requirement
INT-301	KPIs match Business Engine calculations.
INT-302	Comparisons use accurate historical data.
INT-303	Intelligence updates automatically after relevant data changes.
INT-304	No business data can be modified from this module.
INT-305	Recommendations are generated only from configured business rules.

Chapter 13 - Reports

Module Information

Item	Value
Module	Reports
Module ID	RPT
Purpose	Generate, filter, export and print operational and financial reports.
Users	Owner, Manager, Accountant
Access	Read Only

Business Actions

ID	Action	Description
RPT-001	Generate Report	Generate the selected report.
RPT-002	Export Report	Export report in supported formats.
RPT-003	Print Report	Print the generated report.
RPT-004	Apply Filters	Generate reports using selected filters.

Business Rules

ID	Rule
RPT-10 1	Reports shall always use Business Engine data.
RPT-10 2	Reports are read-only and shall never modify business data.
RPT-10 3	Reports shall respect all selected filters.
RPT-10 4	Exported reports shall match the displayed report data.
RPT-10 5	Historical reports shall remain unchanged unless underlying business data is legitimately updated.

Validations

ID	Validation
RPT-201	Report Type is mandatory.
RPT-202	Required filters must be selected before report generation.
RPT-203	Export is permitted only after successful report generation.

Data Sources

Source	Usage
Business Engine	Report Calculations
Batch Details	Batch Reports
Daily Log	Operational Reports
Liftings	Performance Reports
Financial Ledger	Financial Reports
Final Settlement	Settlement Reports

Acceptance Criteria

ID	Requirement
RPT-301	Generated reports match Business Engine calculations.
RPT-302	Reports respect selected filters.
RPT-303	Exported reports match displayed reports.
RPT-304	Reports do not modify business data.
RPT-305	Supported export formats generate successfully.

Chapter 14 - Settings

Module Information

Item	Value
Module	Settings
Module ID	SETG
Purpose	Configure application masters, business parameters and system settings used throughout the application.
Users	Owner, Administrator
Access	Read, Create, Update

Business Actions

ID	Action	Description
SETG-001	Manage Farms	Create and maintain farm information.
SETG-002	Manage Sheds	Create and maintain shed information.

SETG-00 3	Manage Companies	Maintain integration/company details.
SETG-00 4	Manage Users	Create, update and assign user roles.
SETG-00 5	Manage Standard Curves	Configure standard performance benchmarks.
SETG-00 6	Manage Lookup Tables	Maintain configurable lookup values used by the application.
SETG-00 7	Manage Custom Metrics	Configure additional calculated metrics.
SETG-00 8	Manage Master Data	Maintain reusable master records.
SETG-00 9	Backup & Restore	Manage application backup and restore operations.

Business Rules

ID	Rule
SETG-10 1	Changes to Settings shall affect future calculations unless explicitly configured otherwise.
SETG-10 2	Business Engine shall use the latest approved configuration values.
SETG-10 3	Only authorized users may modify Settings.
SETG-10 4	Deleting master data referenced by transactions is not permitted.
SETG-10 5	Configuration changes shall be audit logged.
SETG-10 6	Standard Curves and Custom Metrics shall be configurable without code changes.
SETG-10 7	Backup and Restore operations shall be restricted to authorized users only.

Validations

ID	Validation
SETG-201	Mandatory master fields must be completed before saving.
SETG-202	Duplicate master records are not permitted where uniqueness is required.
SETG-203	Configuration values shall be validated before being applied.
SETG-204	Restore operations require confirmation before execution.

Data Sources

Source	Usage
Business Engine	Configuration & Formula Processing
Database	Master Data
User Management	Roles & Permissions
Audit Log	Configuration History

Acceptance Criteria

ID	Requirement
SETG-301	Configuration changes are available throughout the application.
SETG-302	Business Engine uses updated configuration values correctly.
SETG-303	Referenced master data cannot be deleted.
SETG-304	All configuration changes are audit logged.
SETG-305	Backup and Restore operations complete successfully.

Chapter 15 - Database Design

Database Principles

ID	Rule
----	------

- DB-00 Use a normalized relational database.
1
- DB-00 Avoid duplicate business data wherever possible.
2
- DB-00 Use UUIDs as primary keys unless otherwise justified.
3
- DB-00 Enforce referential integrity using foreign keys.
4
- DB-00 Archive records instead of permanently deleting business data.
5
- DB-00 Every business table shall support audit tracking (Created By, Created On,
6 Modified By, Modified On).
-

Core Entities

Entity	Purpose
Farms	Farm master records.
Sheds	Shed master records.
Batches	Batch master records.
Daily Logs	Daily operational records.
Liftings	Bird lifting transactions.
Financial Ledger	Income and expense transactions.
Settlements	Final settlement records.
Users	User accounts.
Roles	User roles and permissions.
Lookup Tables	Configurable application values.
Standard Curves	Performance benchmark data.
Custom Metrics	User-defined business metrics.
Audit Logs	System audit history.

Entity Relationships

Parent	Child	Relationship
Farm	Shed	One to Many
Shed	Batch	One to Many
Batch	Daily Log	One to Many
Batch	Liftings	One to Many
Batch	Financial Ledger	One to Many
Batch	Settlement	One to One
Role	User	One to Many

Constraints

ID	Constraint
DB-101	One active batch per shed.
DB-102	Batch Number must be unique.
DB-103	One Daily Log per batch per date.
DB-104	Settlement must belong to one batch only.
DB-105	Financial transactions must belong to one batch only.
DB-106	Master records referenced by transactions cannot be deleted.

Audit Fields

Every business entity shall include:

Field
Created By
Created On
Modified By

Modified On

Archived By (if applicable)

Archived On (if applicable)

Acceptance Criteria

ID	Requirement
DB-20 1	All relationships enforce referential integrity.
DB-20 2	Business constraints are enforced by the database and Business Engine where appropriate.
DB-20 3	Historical records remain intact after archive operations.
DB-20 4	Every business record is traceable through audit fields.

Chapter 16 - Business Engine Formula Catalogue

Formula Principles

ID	Rule
FRM-001	All business calculations shall be performed exclusively by the Business Engine.
FRM-002	Formula inputs shall originate only from approved application data sources.
FRM-003	Formula outputs are read-only and shall not be manually edited.
FRM-004	Formula changes shall take effect immediately unless versioning is enabled.
FRM-005	Every calculated value shall be reproducible from its source data.

Standard Formula Catalogue

ID	Formula	Purpose
FRM-101	Mortality	Total bird deaths recorded during the batch.
FRM-102	Mortality %	Percentage mortality based on placed chicks.
FRM-103	Live Birds	Current live bird count.
FRM-104	Feed Consumed	Total feed consumed during the batch.
FRM-105	Feed Remaining	Remaining feed inventory.
FRM-106	Average Daily Feed	Average feed consumed per day.
FRM-107	Water Consumption	Total water consumed.
FRM-108	Average Bird Weight	Average bird weight based on lifting data.
FRM-109	Total Live Weight	Total weight of lifted birds.
FRM-110	Feed Conversion Ratio (FCR)	Feed efficiency calculation.
FRM-111	Production Cost (PC)	Total production cost per kilogram.
FRM-112	Growing Charges (GC)	Company payment based on configured rules.
FRM-113	Total Income	Sum of all income transactions.
FRM-114	Total Expense	Sum of all expense transactions.
FRM-115	Net Profit / Loss	Final financial result of the batch.
FRM-116	Days in House	Total batch duration.
FRM-117	Performance Index	Overall batch performance indicator.

Derived Metrics

ID	Metric	Description
FRM-201	Feed per Bird	Feed consumed per live bird.
FRM-202	Water per Bird	Water consumed per live bird.
FRM-203	Daily Weight Gain	Average weight gain per day.
FRM-204	Cost per Bird	Production cost allocated per bird.

FRM-205	Cost per Kg	Production cost allocated per kilogram.
FRM-206	Revenue per Kg	Income earned per kilogram.
FRM-207	Profit per Bird	Profit generated per bird.
FRM-208	Profit per Kg	Profit generated per kilogram.

Formula Dependencies

Formula	Depends On
Mortality	Daily Log
Live Birds	Batch Details, Daily Log
Feed Consumed	Daily Log
Feed Remaining	Daily Log
Water Consumption	Daily Log
Average Bird Weight	Liftings
Total Live Weight	Liftings
FCR	Feed Consumed, Total Live Weight
Production Cost	Financial Ledger
Growing Charges	Production Cost, Settings
Net Profit / Loss	Income, Expenses, Growing Charges
Performance Index	Business Engine Metrics

Business Rules

ID	Rule
FRM-30 1	Formula execution shall be automatic whenever source data changes.
FRM-30 2	Circular formula dependencies are not permitted.

FRM-30 3	Custom Metrics may reference existing calculated values but shall not overwrite standard formulas.
FRM-30 4	Formula failures shall not modify stored business data.
FRM-30 5	Every calculated value shall identify its source inputs for traceability.

Application Wide Constants

Application Timezone

Date Format

Currency

Weight Unit

Feed Unit

Water Unit

Default Precision

Default Pagination

Default Archive Behaviour

Acceptance Criteria

ID	Requirement
FRM-401	Standard formulas produce consistent results for identical inputs.
FRM-402	Formula recalculation occurs automatically after source data changes.
FRM-403	Derived metrics are calculated correctly from dependent formulas.
FRM-404	Custom Metrics coexist with standard formulas without conflict.
FRM-405	All calculated values are traceable to their source data.

Chapter 17 - Permission Matrix

Permission Principles

ID	Rule
PER-001	Access shall be controlled through role-based access control (RBAC).
PER-002	Users shall only have the minimum permissions required to perform their responsibilities.
PER-003	Permissions shall be evaluated before every business action.
PER-004	Unauthorized actions shall be blocked and audit logged.
PER-005	Roles may be extended in the future without affecting existing permission assignments.

Standard Roles

Role	Purpose
Owner	Full system access.
Administrator	Manage application configuration, users and system settings.
Manager	Manage daily farm operations and batches.
Accountant	Manage financial records and settlements.
Viewer	Read-only access to permitted modules.

Module Permission Matrix

Module	Owner	Administrator	Manager	Accountant	Viewer
Dashboard	Full	Read	Read	Read	Read
Batch Centre	Full	Full	Full	Read	Read

Batch Details	Full	Full	Full	Read	Read
Daily Log	Full	Full	Full	Read	Read
Liftings	Full	Full	Full	Read	Read
Financial Ledger	Full	Full	Read	Full	Read
Final Settlement	Full	Full	Read	Full	Read
Intelligence	Full	Read	Read	Read	Read
Reports	Full	Read	Read	Read	Read
Settings	Full	Full	Read*	Read*	No Access

* Limited to configuration explicitly granted by the Owner or Administrator.

Business Action Permissions

ID	Rule
PER-101	Only authorized users may create business records.
PER-102	Only authorized users may edit business records.
PER-103	Only authorized users may delete business records where deletion is permitted.
PER-104	Settlement approval shall be restricted to Owner and Accountant.
PER-105	Batch archive shall be restricted to Owner and Administrator.
PER-106	User and role management shall be restricted to Owner and Administrator.
PER-107	Backup and Restore shall be restricted to Owner and Administrator.
PER-108	Reopening a finalized settlement requires Owner authorization.

Security Rules

ID	Rule
PER-201	Every business action shall verify user permissions before execution.

- PER-2 Permission changes shall take effect immediately.
02
- PER-2 Permission failures shall not expose protected business data.
03
- PER-2 All denied access attempts shall be audit logged.
04
- PER-2 Business Engine calculations shall execute independently of user permissions,
05 but only authorized users may view or act on the results.
-

Acceptance Criteria

- | ID | Requirement |
|---------|--|
| PER-301 | Users can access only the modules permitted by their role. |
| PER-302 | Unauthorized actions are prevented. |
| PER-303 | Permission changes are applied immediately. |
| PER-304 | Restricted business actions require appropriate authorization. |
| PER-305 | Permission violations are recorded in the audit log. |

Chapter 18 - Audit & Logging

Audit Principles

- | ID | Rule |
|---------|---|
| AUD-001 | Every business-critical action shall be audit logged. |
| AUD-002 | Audit records are immutable and cannot be edited or deleted by users. |
| AUD-003 | Audit logging shall occur automatically without user intervention. |
| AUD-004 | Audit logging shall not interrupt successful business operations. |
| AUD-005 | Audit history shall remain available for archived batches. |
-

Auditable Events

ID	Event
AUD-101	User Login
AUD-102	User Logout
AUD-103	Batch Created
AUD-104	Batch Updated
AUD-105	Batch Archived
AUD-106	Daily Log Created
AUD-107	Daily Log Updated
AUD-108	Lifting Added
AUD-109	Lifting Updated
AUD-110	Financial Transaction Created
AUD-111	Financial Transaction Updated
AUD-112	Financial Transaction Deleted
AUD-113	Settlement Generated
AUD-114	Settlement Approved
AUD-115	Settlement Reopened
AUD-116	Settings Modified
AUD-117	User Permissions Modified
AUD-118	Backup Completed
AUD-119	Restore Completed

Audit Information

Every audit record shall capture the following information:

Field

Event ID

Event Type

Entity

Entity ID

Action Performed

Previous Value (where
applicable)

New Value (where applicable)

User

Timestamp

Source Module

Logging Rules

ID	Rule
AUD-20 1	Business errors shall be recorded in the application log.
AUD-20 2	Validation failures shall not create audit records unless a business action was attempted.
AUD-20 3	Permission denials shall be audit logged.
AUD-20 4	Business Engine calculation failures shall be logged with sufficient diagnostic information.
AUD-20 5	System logs and audit logs shall be maintained separately.
AUD-20 6	Sensitive information shall never be stored in logs in plain text.

Retention Rules

ID	Rule
AUD-301	Audit history shall remain available for the lifetime of the associated business records.

AUD-302 Archived batches shall retain complete audit history.

AUD-303 Audit records shall be searchable and sortable by entity, user and date.

Acceptance Criteria

ID	Requirement
AUD-401	All business-critical actions generate audit records.
AUD-402	Audit records cannot be modified by users.
AUD-403	Permission denials are recorded successfully.
AUD-404	Business errors are written to application logs.
AUD-405	Audit history remains available after batch archival.

Chapter 19 - Testing & Acceptance Checklist

Testing Principles

ID	Rule
TST-001	Every module shall be tested independently before end-to-end testing.
TST-002	Business Engine calculations shall be verified using known test data.
TST-003	All critical business workflows shall pass before production deployment.
TST-004	Testing shall include positive, negative and boundary scenarios.
TST-005	Every resolved defect shall be regression tested.

Functional Testing

ID	Requirement
TST-101	Dashboard displays accurate live information.

TST-102	Batch lifecycle executes successfully from creation to archive.
TST-103	Daily Log records update all dependent calculations correctly.
TST-104	Lifting records produce correct performance calculations.
TST-105	Financial Ledger calculates production costs accurately.
TST-106	Final Settlement produces correct financial results.
TST-107	Intelligence displays accurate KPIs and comparisons.
TST-108	Reports generate correct outputs.
TST-109	Settings changes update Business Engine behaviour where applicable.

Business Rule Testing

ID	Requirement
TST-201	One active batch per shed is enforced.
TST-202	Duplicate daily logs for the same batch and date are prevented.
TST-203	Financial transactions are correctly linked to their batch.
TST-204	Settlement cannot be approved with incomplete business data.
TST-205	Archived batches remain read-only.
TST-206	Business constraints are enforced consistently across all modules.

Security Testing

ID	Requirement
TST-301	Unauthorized users cannot access restricted modules.
TST-302	Unauthorized business actions are prevented.
TST-303	Role changes take effect immediately.
TST-304	Permission denials are audit logged.

Data Integrity Testing

ID	Requirement
TST-401	Business Engine calculations remain consistent after data updates.
TST-402	Referential integrity is maintained across all entities.
TST-403	Archived records remain accessible without modification.
TST-404	Audit history remains intact after business operations.

Performance Testing

ID	Requirement
TST-501	Dashboard loads within acceptable performance limits.
TST-502	Batch operations complete without data loss.
TST-503	Reports generate successfully for large historical datasets.
TST-504	Intelligence calculations complete without application failure.

User Acceptance Criteria

ID	Requirement
TST-601	All functional modules operate according to the approved specification.
TST-602	Business calculations match expected results.
TST-603	Role-based permissions function correctly.
TST-604	Audit logging operates correctly.
TST-605	End-to-end batch lifecycle completes successfully.
TST-606	No Critical or High severity defects remain unresolved.

CHAPTER 20 — Business Operations Catalogue

Purpose

Define every business operation supported by the application, including inputs, outputs, business rules and validation requirements. This catalogue serves as the implementation contract between the User Interface, Business Engine and Database.

Operation Template

Item	Description
Operation ID	Unique business operation identifier
Module	Owning module
Purpose	Business objective
Inputs	Required business data
Outputs	Expected result
Business Rules	Rules governing execution
Validations	Validation requirements
Success	Successful outcome
Failure	Possible business errors

Operations

Batch Centre

ID	Operation	Purpose
OPS-BAT-001	Create Batch	Create a new production batch.
OPS-BAT-002	Update Batch	Modify batch information.
OPS-BAT-003	Archive Batch	Archive completed batch.
OPS-BAT-004	Reopen Batch	Reopen archived batch (authorized users only).

Daily Log

ID	Operation
OPS-DLG-001	Create Daily Log
OPS-DLG-002	Update Daily Log
OPS-DLG-003	Calculate Daily Metrics

Liftings

ID	Operation
OPS-LFT-001	Record Lifting
OPS-LFT-002	Update Lifting
OPS-LFT-003	Complete Liftings

Financial Ledger

ID	Operation
OPS-FIN-001	Add Transaction
OPS-FIN-002	Update Transaction
OPS-FIN-003	Delete Transaction
OPS-FIN-004	Categorize Transaction

Settlement

ID	Operation
OPS-SET-001	Generate Settlement
OPS-SET-002	Approve Settlement
OPS-SET-003	Reopen Settlement

OPS-SET-004	Archive Batch
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Reports

ID	Operation
OPS-RPT-001	Generate Report
OPS-RPT-002	Export Report
OPS-RPT-003	Print Report

Settings

ID	Operation
OPS-STG-001	Manage Masters
OPS-STG-002	Manage Users
OPS-STG-003	Configure Standard Curves
OPS-STG-004	Configure Custom Metrics
OPS-STG-005	Backup Database
OPS-STG-006	Restore Database

Business Rules

ID	Rule
OPS-101	Every business operation shall validate permissions before execution.

OPS-10 2	Every business operation shall enforce business rules before data modification.
OPS-10 3	Successful operations shall generate audit records where applicable.
OPS-10 4	Failed operations shall not leave partial business data.
OPS-10 5	Business Engine calculations shall execute automatically whenever required by an operation.

Acceptance Criteria

ID	Requirement
OPS-20 1	All documented operations are implemented.
OPS-20 2	Operations execute according to defined business rules.
OPS-20 3	Invalid operations are rejected with appropriate business errors.

CHAPTER 21 — Business State Transitions

Purpose

Define every valid lifecycle transition within the application.

Batch Lifecycle

Current State	Next State	Allowed	Trigger
Draft	Ready	Yes	Batch Setup Completed
Ready	Running	Yes	Chicks Placed

Running	Lifting	Yes	First Lifting Recorded
Lifting	Settlement	Yes	All Birds Lifted
Settlement	Archived	Yes	Settlement Approved

Settlement Lifecycle

Current	Next
Draft	Generated
Generated	Approved
Approved	Reopened
Reopened	Approved

Daily Log Lifecycle

Current	Next
Draft	Saved
Saved	Updated

Lifting Lifecycle

Current	Next
Draft	Saved
Saved	Updated
Updated	Completed

User Status

Current	Next
Active	Disabled
Disabled	Active

Business Rules

ID	Rule
STA-101	Invalid state transitions shall be rejected.
STA-102	State transitions shall satisfy prerequisite business conditions.
STA-103	Terminal states shall not permit further processing unless explicitly reopened.
STA-104	Every state transition shall be audit logged.

Acceptance Criteria

ID	Requirement
STA-201	Only valid state transitions are permitted.
STA-202	Invalid transitions are blocked.
STA-203	State history remains traceable.

CHAPTER 22 — Business Error Catalogue

Purpose

Define standardized business errors used throughout the application.

Error Categories

Prefix	Module
BAT	Batch Centre
DLG	Daily Log
LFT	Liftings
FIN	Financial Ledger
SET	Settlement
RPT	Reports
STG	Settings
PER	Permissions
ENG	Business Engine
SYS	System

Standard Errors

Batch

Code	Message
BAT-001	Active batch already exists for this shed.
BAT-002	Batch number already exists.
BAT-003	Batch cannot be archived before settlement approval.
BAT-004	Batch is already archived.

Daily Log

Code	Message
DLG-001	Daily Log already exists for this date.
DLG-002	Invalid mortality value.
DLG-003	Feed quantity cannot be negative.
DLG-004	Water quantity cannot be negative.

Liftings

Code	Message
LFT-001	Lifting date is invalid.
LFT-002	Lifted bird count exceeds live birds.
LFT-003	Total weight is invalid.

Financial Ledger

Code	Message
FIN-001	Invalid transaction amount.
FIN-002	Transaction category is required.
FIN-003	Transaction cannot be modified after settlement approval.

Settlement

Code	Message
SET-001	Settlement cannot be generated until all birds are lifted.
SET-002	Settlement is already approved.
SET-003	Settlement requires authorized approval.

Permissions

Code	Message
PER-001	You do not have permission to perform this action.

Business Engine

Code	Message
------	---------

ENG-001	Formula execution failed.
ENG-002	Required calculation data is missing.
ENG-003	Circular formula dependency detected.

System

Code	Message
SYS-001	Unexpected system error occurred.
SYS-002	Database operation failed.
SYS-003	Audit logging failed.

Business Rules

ID	Rule
ERR-101	Every business error shall have a unique code.
ERR-102	Business errors shall be meaningful and actionable.
ERR-103	Internal technical details shall not be exposed to end users.
ERR-104	All business errors shall be available for localization.
ERR-105	Business errors shall be reused consistently across the application.

Acceptance Criteria

ID	Requirement
ERR-20 1	All business validations return standardized error codes.
ERR-20 2	Error messages are consistent across modules.
ERR-20 3	Business errors support troubleshooting without exposing implementation details.

